Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production full-stack engineering, specializing in **agentic AI systems, RAG pipelines, LLM evaluation, and full-stack AI feature delivery** for real estate and compliance domains. Built AgentProbe — an open-source multi-agent AI framework with a 5-metric LLM-as-judge eval engine, React observability dashboard, CI/CD quality gates, and multi-provider LLM routing (OpenAI, Anthropic, Groq, Ollama) via 15-endpoint FastAPI backend. Built LeaseI — an agentic LangGraph RAG pipeline auditing 1,240 lease PDFs for compliance (93.6% precision, 87.9% F1). Shipped production AI and data pipelines across 1.6M+ financial records and 1M+ property records at Entrata. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Built and shipped Entrata's **first production AI pipeline** (Python, PyTorch, OpenAI API) across 1.6M real financial records — designed output validation, quality gates, and compliance controls; delivered to production; later productized into a customer-facing feature.
- Built **20+ full-stack production modules** (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users across 1M+ property records — 99.5% uptime over 6+ months; improved query performance 30% across 200+ SQL queries.
- Developed **affordable housing compliance automation** (LIHTC/Tax Credit, Section 8/HCV, HUD) — income eligibility, AMI calculations, recertification workflows across 100K+ unit portfolios; eliminated 80% of manual workload; zero compliance violations.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents); cut execution time 50% via Docker containerization; mentored 5+ engineers through 300+ peer code reviews.

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Engineering · Statistical Modeling · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership — improved research comprehension across two cohorts by 40%.

PROJECTS

AgentProbe: Full-Stack Agentic AI Evaluation Framework

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · OpenAI · Anthropic · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built end-to-end: **React observability dashboard** (per-run metrics, regression drift, latency tracking) + **15-endpoint FastAPI backend** orchestrating 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing (OpenAI, Anthropic, Groq, Ollama) — full-stack AI copilot pattern from UI to agent runtime.
- Designed and built a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD quality gates blocking non-compliant agent runs and ChromaDB RAG pipeline tracking retrieval quality across concurrent test suites — offline evaluation with automated regression detection.

LeaseI: Agentic RAG Pipeline for Real Estate Compliance

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an agentic LangGraph RAG pipeline auditing 1,240 residential lease PDFs for compliance gaps across 10 rules, 5 US states + federal standards — **93.6% precision, 82.9% recall, 87.9% F1**; deterministic rules engine keeps every compliance decision fully auditable.
- Implemented pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with regression gate and LLM-as-judge scoring for both accuracy and reasoning quality — same eval discipline required for production GenAI in PropTech.

AI on FHIR: NLP Platform for Regulated Medical Records

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Built a full-stack NLP platform (Flask + React) translating natural language queries to FHIR APIs using LLMs, BERT, and semantic embeddings — **HIPAA-compliant RBAC**, ~60% reduction in retrieval time; demonstrates full-stack AI delivery in a compliance-constrained domain.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS

AI & Agentic Systems: LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, LLM-as-Judge, LLM Evaluation, Prompt Engineering, Guardrails, ChromaDB, Vector Databases, Semantic Search, Embeddings, OpenAI API, Anthropic API, Groq, Ollama, NLP, BERT, spaCy

Full-Stack & Backend: React, FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, MySQL, SQLite, Schema Design, SQL Optimization, NumPy, Pandas, PyTorch, Scikit-learn, Tableau, Power BI

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), Playwright, Microservices, ETL, Data Pipelines, RabbitMQ, Git, Linux, Agile/Scrum (120+ sprints)